



Chapter 5

MySQL

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Building a website that displays data from a database requires:

- A database management program like MySQL
- A server-side scripting language like PHP
- Using SQL to manipulate data
- Using HTML/CSS for display

- Relational database management system (RDBMS) is widely used.
- Supports SQL standard.
- Open-source, free, and popular software.
- Fast, reliable, scalable, and easy to use.
- PHP + MySQL: Dynamic web applications running on various platforms.
- Suitable for applications of all sizes.
- MySQL is developed, distributed, and supported by Oracle Corporation.

SQL is the standard language for handling relational databases.

SQL is used to insert, search, update, and delete database records in MySQL.

```
CREATE DATABASE databasename;
```

```
CREATE TABLE table_name (  
    column1 datatype constraint,  
    column2 datatype constraint,  
    column3 datatype constraint,  
    ....  
);
```

NOT NULL / UNIQUE / PRIMARY KEY / FOREIGN KEY /CHECK /DEFAULT



```
SELECT * FROM table_name;
```

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

```
DELETE FROM table_name WHERE condition;
```

Details about SQL used in MySQL

https://www.w3schools.com/mysql/mysql_sql.asp

phpMyAdmin

- a free, open-source, web-based PHP tool for easily managing MySQL and MariaDB databases through a web browser
- allowing users to create/delete databases, tables, users, run SQL queries
- perform backups/imports without complex command-line interfaces
- XAMPP is a free, open-source software package that creates a local web server environment (Apache, MariaDB/MySQL, PHP, Perl)
- it includes **phpMyAdmin**

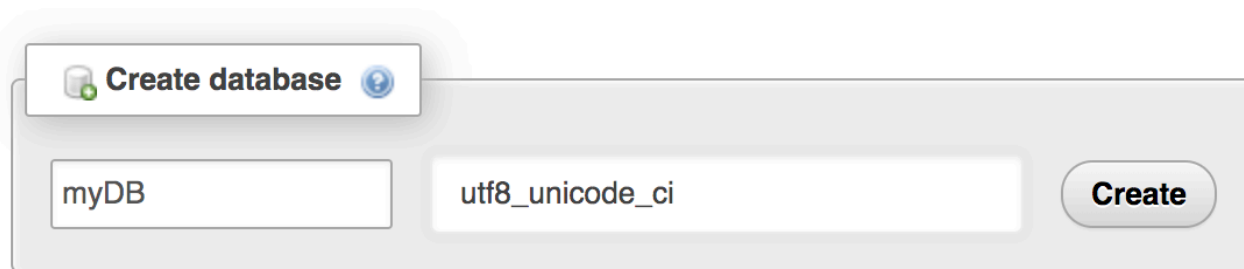
The screenshot shows the phpMyAdmin interface in a web browser. The browser's address bar displays the URL: `localhost/phpmyadmin/index.php?route=/server/databases&lang=en`. The interface has a sidebar on the left with a tree view of databases: `information_schema`, `mysql`, `performance_schema`, `phpmyadmin`, and `test`. The main content area is titled "Databases" and features a "Create database" form with a "Database name" input field containing `utf8mb4_general_ci` and a "Create" button. Below the form are checkboxes for "Check all" and "Drop", and a search bar. A table lists the existing databases with their collations and actions.

Database	Collation	Action
☐ <code>information_schema</code>	<code>utf8_general_ci</code>	Check privileges
☐ <code>mysql</code>	<code>utf8mb4_general_ci</code>	Check privileges
☐ <code>performance_schema</code>	<code>utf8_general_ci</code>	Check privileges
☐ <code>phpmyadmin</code>	<code>utf8_bin</code>	Check privileges
☐ <code>test</code>	<code>utf8mb4_general_ci</code>	Check privileges

Total: 5

- Create a database
- Create a table
- Insert data into a table
- View data in a table
- Execute SQL queries
- Export the database
- Import the database

Create database



myDB utf8_unicode_ci Create

Database names must

- not contain accents, spaces, or special characters
- not be the same as existing database names.

To enable the database to store Vietnamese characters, the utf8_unicode_ci character encoding should be selected.

Create Table/ Insert into

 Browse

 Structure

 SQL

 Search

 Insert

 Export

 Import

Run SQL query/queries on table **myDB.Persons**: 

Execute SQL queries

```
1 SELECT * FROM Persons;
```

PersonID	LastName	FirstName	Address	City
1	Smith	John	123 Maple Street	New York
2	Doe	Jane	456 Oak Avenue	Los Angeles
3	Brown	Charlie	789 Pine Road	Chicago

Export data

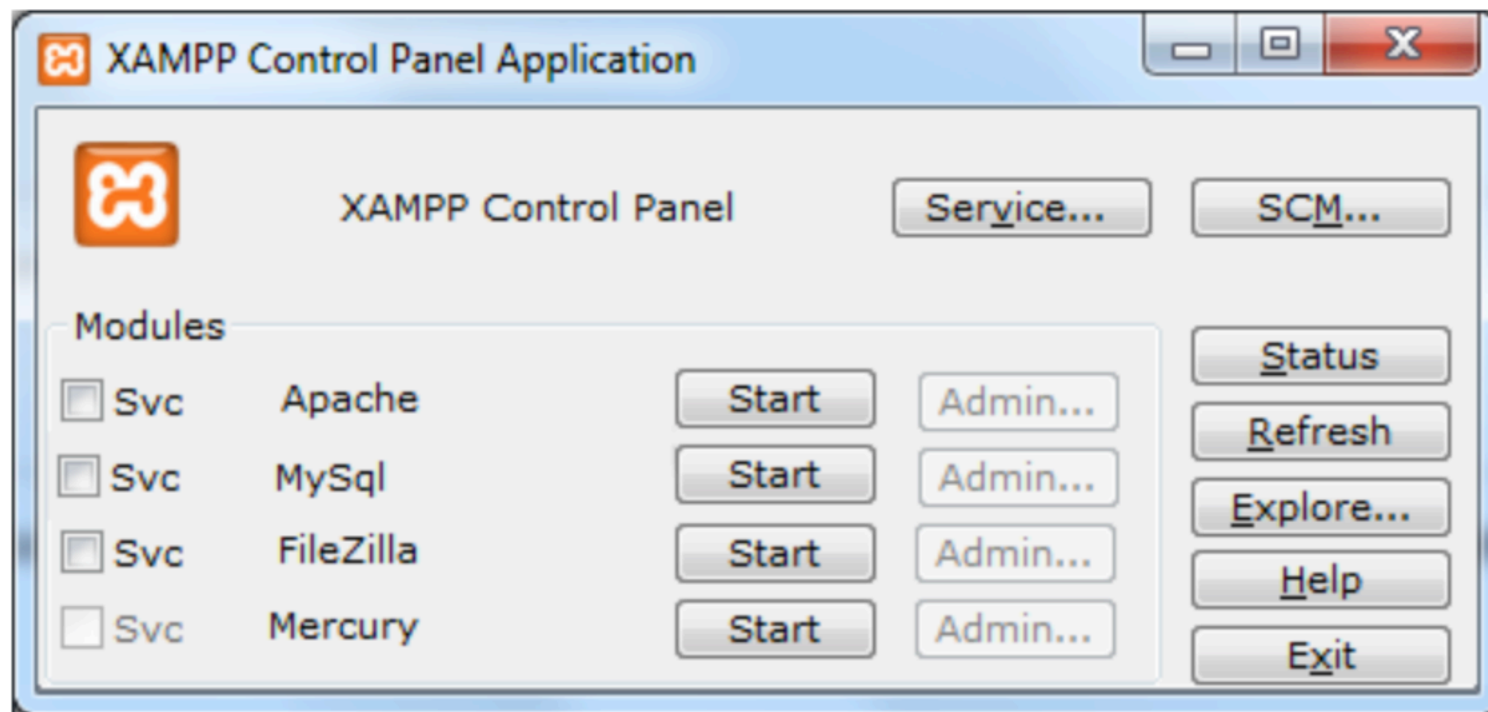
- Use the Export function from the menu.
- Data can be exported in multiple formats.
- This is used for database backup, to prevent potential risks.

Import data

The screenshot shows the phpMyAdmin web interface in a browser. The address bar shows the URL: `localhost/phpmyadmin/index.php?route=/database/import&db=test`. The left sidebar shows a tree view of databases, with 'test' selected. The main content area is titled 'Importing into the database "test"'. It contains two main sections: 'File to import:' and 'Partial import:'. The 'File to import:' section has a text input field with 'Persons.sql' and a 'Choose File' button. Below it, there is a dropdown menu for 'Character set of the file:' set to 'utf-8'. The 'Partial import:' section has a checkbox labeled 'Allow the interruption of an import in case the script detects it is close to the PHP timeout limit.' which is checked. Below this, there is a text input field for 'Skip this number of queries (for SQL) starting from the first one:' with the value '0'. At the bottom, there is a 'Console' tab.

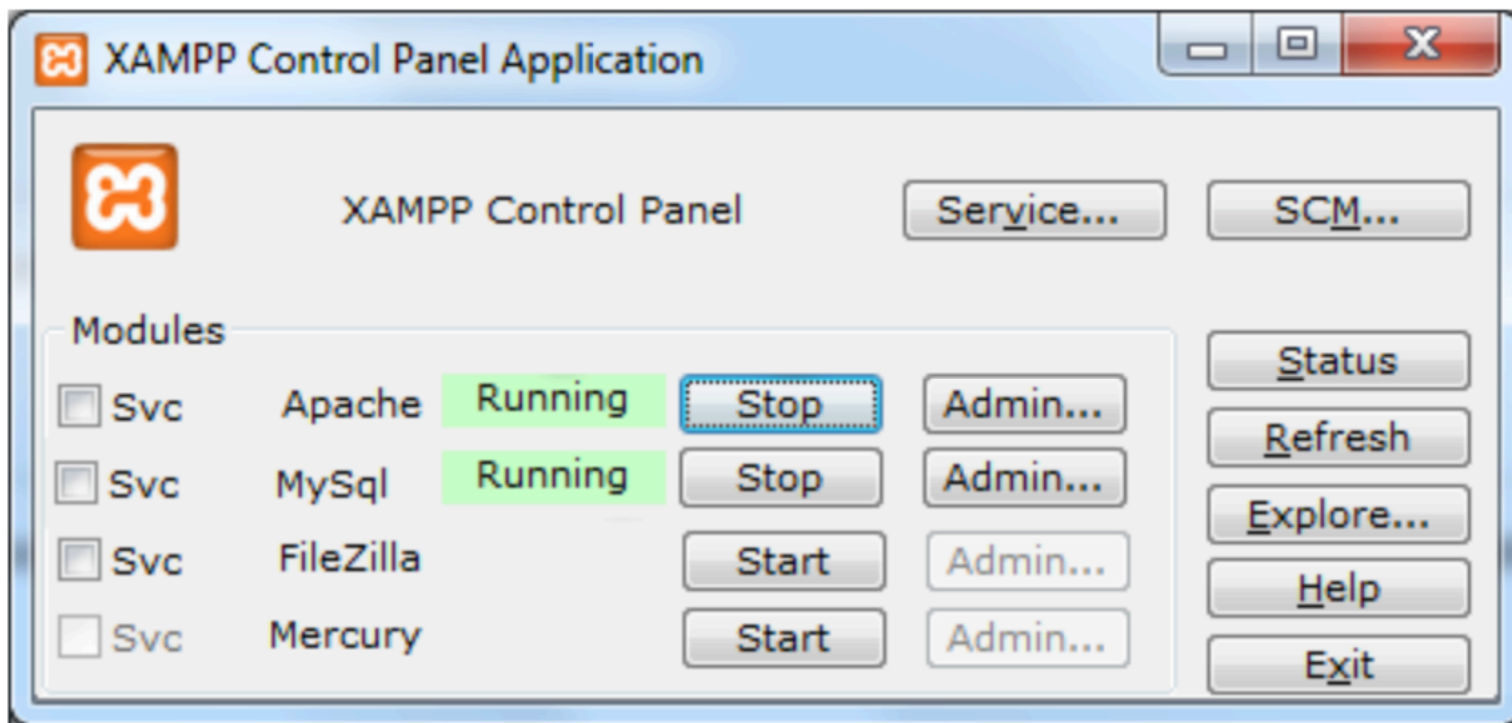
- A free and open source cross-platform web server solution stack package
- Consisting mainly of the Apache HTTP server, MySQL database, and interpreters for scripts written in the PHP and Perl programming languages
- XAMPP's name is an acronym for:
 - X (meaning cross platform)
 - Apache HTTP server
 - MySQL
 - PHP
 - Perl

- Run XAMPP and start Apache and MySQL

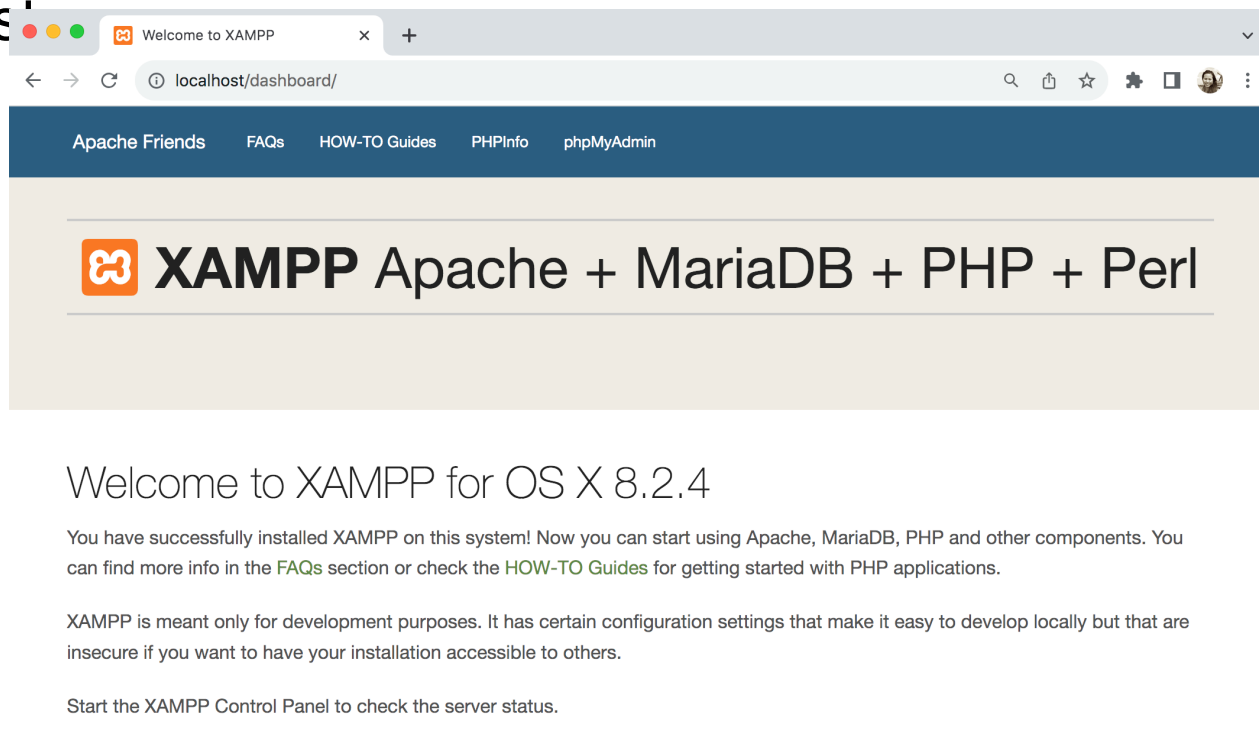


XAMPP

- Click on the Start buttons to start MySQL and Apache services



- Check if Apache is working: open the browser and go to `http://localhost`



- Check if MySQL is running
 - Open the command prompt and run the command: `mysql`
 - If it returns “Command not found”, check whether the current folder is the MySQL folder. If not, change the current folder and rerun the command.
 - If the command returns MySQL information, MySQL is working properly.
 - You can run some other commands to make further checks:
`select version(); select user(); show databases;`
 - Run `exit` to stop

◆ Exercise – Create database and tables

Create database:

```
CREATE DATABASE simple_library;  
USE simple_library;
```

Create members table:

```
CREATE TABLE members (  
    member_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    email VARCHAR(100),  
    join_date DATE,  
    membership_type ENUM('basic', 'premium')  
);
```

Exercise – Create database and tables

Create books table:

```
CREATE TABLE books (  
  book_id INT PRIMARY KEY AUTO_INCREMENT,  
  title VARCHAR(150),  
  author VARCHAR(100),  
  genre VARCHAR(50),  
  price DECIMAL(6,2),  
  available_copies INT );
```

Insert [sample data](#)

Create borrowings table:

```
CREATE TABLE borrowings (  
  borrowing_id INT PRIMARY KEY AUTO_INCREMENT,  
  member_id INT,  
  book_id INT,  
  borrow_date DATE,  
  return_date DATE,  
  status ENUM('borrowed', 'returned', 'overdue'),  
  FOREIGN KEY (member_id) REFERENCES members(member_id),  
  FOREIGN KEY (book_id) REFERENCES books(book_id) );
```

Exercise – Queries

- Q1: List all books in the Fiction genre
- Q2: Find all premium members
- Q3: List books sorted by price (highest first)
- Q4: Count how many books are in each genre
- Q5: Find the most expensive book

◆ Exercise – Queries

- Q6: Show all borrowings with member names and book titles
- Q7: Find all currently borrowed books with member names and book titles
- Q8: List members who have overdue books
- Q9: Show all books borrowed by Carol White
- Q10: Count how many books each member has borrowed

Exercise – Queries

- Q11: Find the most popular books (times borrowed)
- Q12: Calculate total value of books borrowed by each member
- Q13: Find members who have borrowed more than 2 books
- Q14: Find books that have never been borrowed
- Q15: Update a book's status when returned



Questions?